

LASER TRIANGULATION BASED HEIGHT MEASUREMENT

Authors: Mayank Bisht, Suseendhar T.G., Swarnabha, Shashank Shanbhang

2nd October 2026

Abstract

Laser triangulation has emerged as a widely used non-contact technique for surface height measurement in optical metrology. In this project, a single-camera laser triangulation system is developed to measure the height variation of an object relative to a reference plane. It is based on the principle of diffuse reflection, in which incident laser light is scattered from the surface over a range of directions, allowing the illuminated spot to be detected by a camera. A diode laser is used to illuminate the object, while a Raspberry Pi camera captures the position of the resulting laser spot. A change in the height of the object causes a corresponding lateral displacement of the laser spot in the camera's image plane. This pixel displacement is subsequently related to the physical height of the object using a mathematical calibration model. Since the technique is non-contact, it enables surface measurements without mechanically interacting with or altering the object, making it suitable for applications requiring rapid and minimally invasive dimensional characterization. The developed system demonstrates the feasibility of using a low-cost laser, camera, and image-processing-based approach for optical surface height measurement.

1 System Specifications

Table 1: System specifications

Specification	Value
Camera	Raspberry Pi Camera Rev 1 / OV5647
Image resolution used by code(N_x by N_y)	2592×1944 pixels
Pixel pitch	$1.4 \mu\text{m}$
Horizontal FOV(ϕ_{max}^x)	54°
Vertical FOV(ϕ_{max}^y)	41°
Camera-laser baseline(B)	117.3 mm
Camera-to-reference working distance(H)	139 mm
Triangulation angle(θ)	49.9°

The calibration JSON confirms the 2592×1944 camera resolution. The field of view, baseline, working distance and pixel pitch are the project setup values used for the theoretical calculations.

2 Physics behind triangulation

The laser is kept at an angle θ with respect to the camera plane. B is set-up according to which we set H. The relationship between θ , B and H when used in tandem with camera specifications determine the parameters(resolution, field of view and the spot size) of the set-up that are calculated using the equations given below.

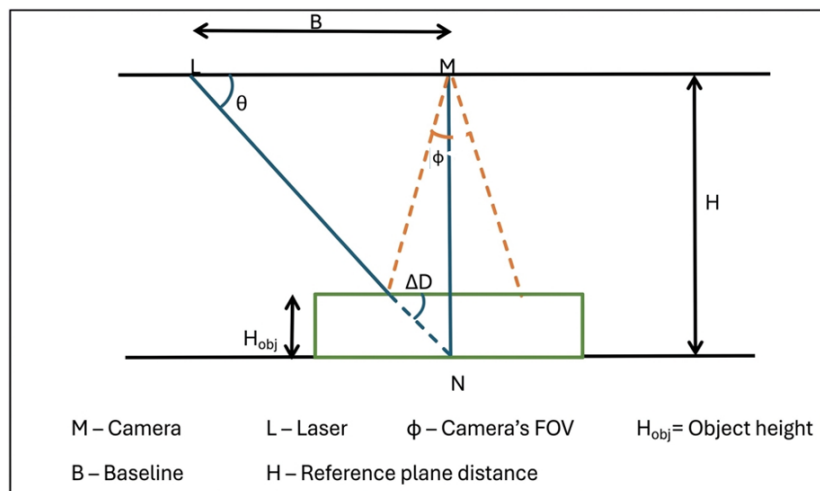


Figure 1: A schematic of laser triangulation.

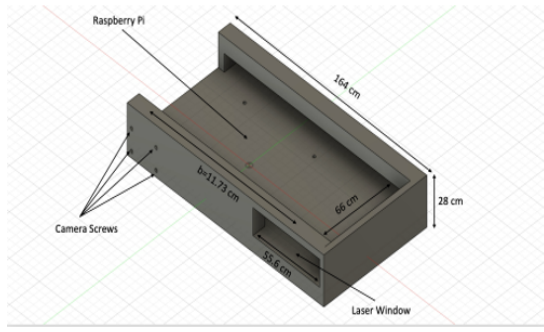
When an object is placed on the reference plane, the camera observes a horizontal shift in the spot position (ΔD) which corresponds to the object height H_{obj} .

$$H_{obj} = \Delta D \cdot \tan \theta \quad (1)$$

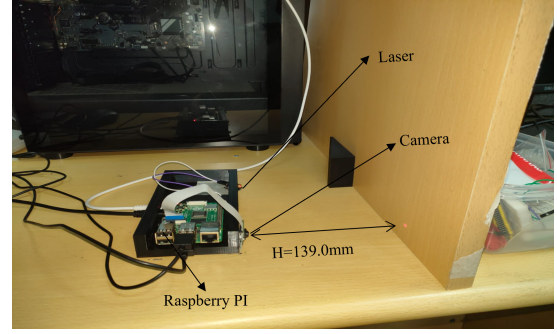
and then ΔD corresponds to the pixel shift (ΔX) in the camera that is used to calculate the height of the object.

3 Set-up Holder

Even minor disturbances can displace the laser spot, which can result in erroneous readings. Hence, the set-up is held in its position by the holder as shown in Figure 2(a).



(a)



(b)

Figure 2: (a) is the figure of holder developed and (b) is the set-up done to perform height measurements.

According to the values of θ and B , H is taken to be 139.00 mm as shown in the figure 2(b).

4 Estimating theoretical parameters

The lateral resolution of the camera tells us the correspondence between the actual lateral distance covered by the spot and corresponding pixel shift occurring in the camera. This is calculated using the following formula

$$R_x = \frac{2B \cdot \tan(\phi_{max}^x/2)}{N_x} \quad (2)$$

R_x for our system is 0.0562 mm/pixel.

Similarly for the the resolution along the chosen y-direction, the formula is

$$R_y = \frac{2B \cdot \tan(\phi_{max}^y/2)}{N_y} \quad (3)$$

R_y for our system is 0.0550 mm/pixel.

The depth sensitivity, given by the equation

$$R_z = R_x \cdot \tan \theta \quad (4)$$

is 0.0670 mm/pixel.

The maximum height possible is given by

$$H_{max} = \frac{B \tan^2(\theta) \tan(\phi_{max}^x/2)}{1 + \tan^2(\theta) \tan(\phi_{max}^x/2)} \quad (5)$$

which gives the height to be 52.15 mm.

5 Basic Working Principle

1. Use the 0 mm position of the laser spot as the reference.
2. Place objects of known height in the measurement setup.

3. Detect the laser spot in each camera frame.
4. Determine the horizontal laser position at every known height.
5. Fit several possible height-versus-pixel-shift models to the calibration data.
6. Select the model that gives the lowest validation error when each calibration point is temporarily left out.
7. Save the selected model and all calibration information in `calibration_model.json`.
8. The height-calculation program loads that JSON file, detects the laser spot on an unknown object, calculates its current pixel shift, and evaluates the selected model to obtain the object height.

The height-calculation program does not create a new curve every time an object is measured. The curve is fitted during calibration and saved in the calibration JSON. The measurement program simply loads the already-selected model and uses it.

6 Calibration Code — Detailed Algorithm

The calibration program is responsible for converting a physical reference height into measured image data. It is the stage where the system gives the relationship between the pixel shift and the object height.

6.1 Camera Setup

The camera is configured with 2592×1944 pixels. The code initially allows the camera exposure and gain to settle, then locks the exposure, analogue gain and colour gains. These settings are stored in the calibration JSON so that the height-calculation program can reproduce the same camera conditions.

6.2 Calibration Heights

The supplied calibration program uses two sets of known heights:

0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30 mm

followed by anchor heights of

34, 40, 46, 50 mm.

Thirty valid measurements are targeted at each height.

6.3 Laser Spot Detection

Stage 1 — Coarse Detection

To ensure that the program examines only the detector region of interest(ROI). A pixel is considered part of the red laser region when the red channel is above 80 and is at least 40 counts above green and 30 counts above blue. A 5×5 morphological opening removes small isolated noise after which the largest remaining contour is selected.

Stage 2 — Fine Centre Calculation

The centroid of the largest contour gives a coarse spot position. A ± 60 -pixel window around this position is then examined using the red channel. The window is Gaussian-blurred with $\sigma = 2$ pixels and the local background is estimated from the blurred 10th percentile pixels.

The brightest point is found, and only pixels above threshold (T) 60% of the peak intensity(P) above the background (B_g) value are retained.

$$T = B_g + 0.6 \cdot P \quad (6)$$

Only the component containing the peak is kept.

The laser centre is a weighted centroid in which the weight is based on the amount by which each selected pixel exceeds the local threshold. This method is advantageous because it gives a sub-pixel estimate of the laser position.

The same process also gives the detected laser beam width and whether the red channel is saturated.

6.4 Repeated Measurements and Averaging

For every known height, the program collects repeated measurements. A median-based inlier test removes occasional abnormal X measurements. The remaining measurements are used to calculate mean X, mean Y, standard deviation and average beam width.

6.5 Creating the Pixel-Shift Data

The 0 mm measurement becomes the reference. For every calibration height, the horizontal shift is calculated as

$$\Delta X = X_{\text{height}} - X_{0\text{ mm}}.$$

The calibration therefore uses the horizontal pixel shift, not the absolute camera X-position. Hence, the measurement program requires a fresh 0 mm reference before measuring objects.

This workflow is explained using a flow chart given in figure 3

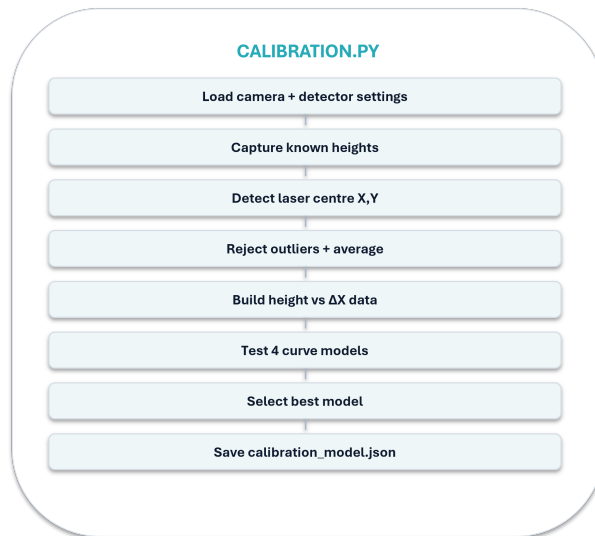


Figure 3: A flow chart explaining calibration process.

7 Curve Fitting and Model Selection

The current calibration code has `MODEL_CHOICE = "auto"`. Therefore, the model is automatically selected during calibration. The four candidate models in the supplied code are Piecewise Cubic Hermite Interpolating Polynomial(PCHIP), rational, polynomial 3 and polynomial 4.

7.1 The Four Curves Used by the Code

7.1.1 PCHIP

PCHIP is a piecewise cubic, monotonic interpolation. It stores the calibration pixel shift (ΔX) and height points(h) as knots. Then it creates a smooth curve between them by generating a cubic expression in the region between two knots and equating each equation and it's first order derivatives at knots. Since, it interpolates the calibration points, its in-sample error can be zero.

7.1.2 Rational Model

The code explicitly uses

$$h = \frac{au + b}{cu + 1}, \quad u = \frac{\Delta X}{1000}$$

where a, b, c are fitted constants.

7.1.3 Polynomial 3

A cubic polynomial is fitted in the normalized variable u :

$$h = c_3u^3 + c_2u^2 + c_1u + c_0.$$

7.1.4 Polynomial 4

A fourth-order polynomial is fitted as

$$h = c_4u^4 + c_3u^3 + c_2u^2 + c_1u + c_0.$$

7.2 Model Selection

The calibration code first fits every candidate model that has enough calibration points. Then it performs a validation test by temporarily removing each interior calibration point, the model is refitted using all the other points, and the removed point is predicted. The difference between the predicted and actual height is recorded. This process is repeated for all internal points.

The root-mean-square value of these leave-one-out errors is then calculated. When `MODEL_CHOICE = "auto"`, the model with the smallest leave-one-out error is selected.

Table 2: Candidate model performance

Model	In-sample (mm)	LOO (mm)	Max LOO (mm)
PCHIP	0.0000	0.3316	0.5203
Rational	0.2103	0.2447	0.5098
Polynomial 3	0.1972	0.2509	0.5012
Polynomial 4	0.1882	0.2777	0.5049

The rational model is selected as it has the lowest leave-one-out error of approximately 0.2447 mm. The rational model was therefore written to `calibration_model.json`.

8 Selected Calibration Model — Exact Equation Used

The selected model stored in the supplied `calibration_model.json` is rational. Its fitted parameters are

Table 3: Stored rational-model parameters

Parameter	Stored value
a	69.943856813482
b	-0.348077262335
c	0.529911405720
X (minimum)	0.000000 px shift
X (maximum)	1156.220310 px shift

The code first normalizes the horizontal shift using

$$u = \frac{\Delta X - X_{\text{CENTER}}}{X_{\text{SCALE}}}.$$

For the current calibration file,

$$X_{\text{CENTER}} = 0, \quad X_{\text{SCALE}} = 1000,$$

so

$$u = \frac{\Delta X}{1000}.$$

Substituting the stored constants gives the exact model used by the height-calculation program:

$$h = \frac{69.9438568135u - 0.3480772623}{0.5299114057u + 1}$$

or directly in terms of pixel shift:

$$h = \frac{0.0699438568\Delta X - 0.3480772623}{0.0005299114\Delta X + 1}.$$

The equation given above converts the measured horizontal pixel shift into height. The height-calculation code does not refit these constants during object measurement; it reads them from `calibration_model.json`.

The calibration JSON also stores the calibrated height range 0–50 mm and horizontal shift range 0–1156.22 pixels. The range is taken to be less than maximum possible limit of 52.15 mm because at the higher heights the lens of the camera causes distortions which gives inaccurate height measurements.

9 Observed System Specifications

Using the camera, we measure some parameters associated with our laser system.

Table 4: Calculated model specifications

Specification	Value
Laser spot starting X-position	1187.92 px
Laser spot ending X-position	2344.15 px
Horizontal laser shift over 0–50 mm	1156.22 px
Vertical shift over 0–50 mm	9.11 px
Detector ROI	$X = 900\text{--}2400$, $Y = 1250\text{--}1550$

10 Height Calculation Code — Detailed Algorithm

After calibration is complete, the supplied height-calculation program performs the actual measurement of unknown objects. It uses the calibration file produced by the calibration program.

10.1 Load the Calibration File

The program opens `calibration_model.json`. It checks that the detector version matches, checks that a reference exists and a selected model exists. It then loads the detector settings, camera settings, reference information, and selected model.

10.2 Reproduce the Calibration Camera Conditions

The camera is configured using the resolution, exposure, analogue gain and colour gains stored in the calibration JSON. This keeps the measurement conditions consistent with calibration.

10.3 Measure the Reference

Before measuring an object height, the 0 mm reference is captured using 30 shots. Outliers in X are rejected and the remaining X values are averaged. This current reference is important because a small common movement of the camera/laser setup can otherwise appear as object height.

10.4 Measure the Object Height

For each object, 20 laser-spot measurements are collected. The exact same detector used during calibration is applied. The X and Y values are filtered and averaged, and the X standard deviation is calculated.

10.5 Convert Laser Position to Pixel Shift

The height calculation uses the formula

$$\Delta X = X_{\text{object}} - X_{\text{reference}}$$

Only ΔX is used for height. The Y shift is recorded for diagnostic purposes but is not the input to the calibration model.

10.6 Height Calculation

The measured ΔX is passed into the stored rational model. The model returns the calculated height in millimeters.

10.7 Range Check

If the measured pixel shift is within the saved pixel range of 0–1156.22, the rational curve is evaluated directly. If it falls outside the calibrated range, the code performs a linear extrapolation using the slope at the nearest calibration edge and marks the result as extrapolated.

10.8 Measurement Uncertainty

The code estimates a repeatability contribution from the measured X standard deviation and the local slope of the calibration curve. Then it combines this with the calibration validation error.

In the results, the uncertainty is approximately 0.245 mm for the five measured objects.

10.9 Saving the Result

The final CSV stores calculated height, uncertainty, mean X/Y, X/Y standard deviation, X/Y shift, beam width, valid shots, rejected outliers and whether extrapolation was used.

11 Pixel Shift Calibration Table

Table 5: Pixel-shift calibration data

Height (mm)	Mean X (px)	Mean Y (px)	ΔX (px)	ΔY (px)	Beam width (px)
0	1187.92	1419.22	0.00	+0.00	105.33
3	1242.71	1417.44	54.79	-1.78	67.73
6	1283.94	1418.09	96.02	-1.13	75.30

Height (mm)	Mean X (px)	Mean Y (px)	ΔX (px)	ΔY (px)	Beam width (px)
9	1333.65	1419.08	145.72	-0.14	82.30
12	1377.70	1420.79	189.78	+1.57	84.37
15	1440.36	1419.23	252.43	+0.01	79.97
18	1488.95	1421.92	301.03	+2.70	82.53
21	1551.40	1422.16	363.48	+2.94	90.50
24	1613.12	1423.02	425.20	+3.80	92.18
27	1678.99	1424.01	491.07	+4.79	98.10
30	1741.49	1425.02	553.57	+5.80	97.73
34	1849.84	1425.19	661.91	+5.97	95.97
40	2024.87	1425.41	836.95	+6.19	103.27
46	2204.26	1429.21	1016.33	+9.99	105.07
50	2344.15	1428.33	1156.22	+9.11	110.43

11.1 Sensitivity

The average sensitivity over the full calibration range is

$$\frac{\text{total horizontal shift (in px)}}{\text{total height change (in mm)}} = \frac{1156.22}{50} = 23.12 \text{ px/mm}$$

and equivalently

$$\frac{1}{\text{sensitivity}} = \frac{50}{1156.22} = 0.04324 \text{ mm/px.}$$

For a 3 mm height change, the laser moves 54.79 pixels, corresponding to

$$\frac{54.79}{3} = 18.26 \text{ px/mm,}$$

or approximately

$$0.05476 \text{ mm/px}$$

over the 0–3 mm interval.

This shows that the sensitivity is a function of the height of the object.

12 Theoretical vs Practical Performance

Table 6: A comparison of theoretical values and practical performance

Parameter	Definition / formula	Theoretical	Practical
Pixel depth / depth resolution	Relation between the object height and corresponding pixel shift in the camera	0.0670 mm/px	0.0432 mm/px
Lateral resolution	Relation between horizontal shift and corresponding pixel shift in the camera	0.0562 mm/px	0.0371 mm from mean X repeatability
Measurement range	Height interval over which system is calibrated / usable	52.15 mm	0–50 mm calibrated
Spatial resolution / laser spot footprint	Detected laser-spot width $\times$ object-plane mm/px	1 pixel reference $\approx$ 0.0562 mm	Average detected spot width $\approx$ 91.4 px $\rightarrow$ 5.14 mm
Height accuracy indicator	Calibration prediction error	—	0.2447 mm Maximum Error = 0.5098 mm

The value obtained from the 0–50 mm calibration shift quantifies how consistently the system detects the same spot centre across repeated measurements. It is a measure of repeatability, not of the system’s ability to resolve two distinct features separated by that distance.

The spatial value is derived from the mean laser-spot width as observed by the detector. It reflects the effective illuminated footprint on the target surface, which is a combination of the laser source width, the optical blur, and the detector pixel size. It should not be interpreted as an independently measured optical point-spread function.

13 Measurement Results and Accuracy

The supplied `height_measurement_results.csv` contains five object measurements.

Table 7: Measured object heights

Object	Calculated height (mm)	Uncertainty (mm)	Mean X (px)	ΔX (px)
1	27.1900	0.2447	1680.5509	495.8638
2	0.6351	0.2447	1198.8117	14.1246
3	10.5186	0.2448	1353.5031	168.8160
4	24.1059	0.2447	1612.4301	427.7430
5	10.3304	0.2447	1350.3220	165.6349

The supplied measurement-results CSV does not contain the true physical height of each of these five objects. Therefore, a true-object accuracy calculation given by

$$|h_{\text{measured}} - h_{\text{actual}}|$$

cannot be calculated from this CSV without assuming the object heights. Hence, no such assumption has been made in this report.

The selected rational model has a leave-one-out RMS error of 0.2447 mm and a maximum leave-one-out error of 0.5098 mm.

Relative to the 50 mm calibrated range, these correspond to approximately 0.49% RMS of full scale and 1.02 % maximum of full scale.

14 Files Produced by the Calibration and Measurement System

Table 8: Files produced by the system

File	Purpose
<code>all_measurements.csv</code>	Stores individual laser-spot measurements captured during calibration.
<code>combined_calibration_data.csv</code>	Stores averaged calibration points: height, mean X/Y, shifts, spread and beam width.
<code>calibration_analysis.csv</code>	Stores validation errors for each candidate model and the selected model's in-sample errors.
<code>calibration_coefficients.csv</code>	Stores candidate models, fitted parameters and validation errors.
<code>calibration_model.json</code>	Final calibration package read by <code>height_calc.py</code> . Contains detector settings, camera settings, reference, calibration points, selected model and validation error.
<code>calibration_fit_comparison.png</code>	Plot comparing the four candidate calibration curves.
<code>height_vs_horizontal_pixels.png</code>	Diagnostic plot of height versus absolute laser X-position.
<code>calibration_error_comparison.png</code>	Leave-one-out error comparison between candidate models.
<code>pixel_position_standard_deviation.png</code>	Shows X and Y spot-position stability versus height.
<code>calibration_fit_residuals.png</code>	Shows residuals of the selected rational model.
<code>height_measurement_results.csv</code>	Stores final unknown-object height measurements and associated measurement information.

14.1 Relationship Between the Two Main Programs

The central data path of the project is

`calibration(2).py` → calibration CSV files + model comparison →
`calibration_model.json` → height

`calc(1).py` → `height_measurement_results.csv`

The calibration program creates the information needed by the height-calculation program; the height-calculation program does not independently rediscover the calibration relationship.

15 Final Project Summary

- **Objective:** To measure object height using the laser triangulation method.
- **Calibration Code:** It captures known heights, detects the laser spot, averages repeated measurements, calculates the shift from the 0 mm reference, fits four possible height-versus-shift models and automatically selects the model with the smallest leave-one-out validation error.
- **Optimum model choice:** The calibration program is in automatic model-selection mode. The rational model gave the smallest leave-one-out validation error of 0.2447 mm, so it was written to `calibration_model.json`.
- **Height calculation:** The saved calibration JSON is loaded that reproduces the calibration camera/detector settings, measures a fresh 0 mm reference, measures the object, calculates ΔX , evaluates the stored rational equation and writes the calculated height and uncertainty to a CSV file.
- **Calibration result:** The laser spot moves from $X = 1187.92$ pixels at 0 mm to $X = 2344.15$ pixels at 50 mm, giving a total horizontal shift of 1156.22 pixels, or an average of 23.12 pixels/mm.
- **Calibration validation:** The selected model gives 0.2447 mm leave-one-out RMS error and 0.5098 mm maximum leave-one-out error over the calibration data.
- **Measurement result:** The current measurement CSV contains five calculated object heights, but it does not store their true physical heights. Therefore an independent object-accuracy percentage cannot be calculated from that file without additional true-height data.

16 Source Files Used

The report was prepared from the supplied project files, principally:

- `calibration(2).py`
- `height_calc(1).py`
- `calibration_model.json`
- `combined_calibration_data(1).csv`

- `calibration_analysis(2).csv`
- `calibration_coefficients(1).csv`
- `height_measurement_results.csv`